

So initial velocity is negative maximum, so answer D.

12.

Ans: A

Distance between the two points along the wave (perpendicular to wavefront)

$$= 8.0 \sin 30^\circ$$

$$\text{phase difference } \phi = \frac{x}{\lambda} \times 360^\circ = \frac{8.0 \sin 30^\circ}{40} \times 360^\circ = 36^\circ$$

12

Ans: C